

Krishav Sambyal



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Professional Summary

Automation engineer with 3+ years of experience at Capgemini India in enterprise infrastructure and workflow automation. Currently working on the deployment of a distributed Ansible Automation Platform (AAP) integrating ITSM (ServiceNow), observability (Splunk), iPaaS and secrets management — a closed-loop, event-driven automation system operating across customer environments. Strong interest in the design, efficiency, reliability and security of complex computing systems. Seeking to transition from applied engineering into research through an MS, building rigorous theoretical foundations to contribute meaningfully to systems research.

Research Interests

- Artificial intelligence and agentic systems — large language models, autonomous agents and AI-driven decision-making
- Intelligent and autonomous systems — applying AI and machine learning to system monitoring, decision-making and self-healing behaviour
- Reliability and verification of software systems — static analysis, correctness and dependability
- Networked and connected systems — efficiency, scalability and reliability in real-world environments
- Systems and network security — secure design, access control and vulnerability analysis
- Cyber-physical and embedded systems

Education

Bachelor of Engineering — Computer Science and Engineering

University Institute of Engineering, Chandigarh University, India | 2020

Projects

AnsibleGuard — Static Analysis Tool for Ansible Playbooks (Python, Open Source)

- Open-source tool performing rule-based static analysis on infrastructure-as-code, detecting security vulnerabilities, non-idempotent commands and best-practice violations before deployment
- Modular check engine with three rule categories (security, best-practice, style), severity classification, CLI interface and JSON output for CI/CD pipeline integration; unit-tested
- Explores the application of program analysis techniques to infrastructure-as-code correctness — an area of growing importance as organisations scale automation

PyAutoHeal — Event-Driven Auto-Remediation System (Python) [In Progress]

- Prototype closed-loop remediation engine: monitors system metrics, detects threshold breaches, autonomously restarts affected services, verifies recovery and logs outcomes
- Explores self-healing behaviour in computing systems — a small-scale investigation of the detect-act-verify loop used in autonomous infrastructure recovery

UAV-Based Wireless Security Experimentation (Undergraduate Final Year Project)

- Designed and implemented a custom quadcopter UAV platform, integrating flight controller, motors, ESCs and avionics components

- Developed an ESP8266-based wireless experimentation module to study IEEE 802.11 management frame behaviour in controlled academic environments
- Implemented embedded firmware and a web-based control interface to conduct controlled de-authentication and beacon frame experiments for analysing network resilience
- Applied concepts of wireless communication, embedded systems, computer networks and cyber security to evaluate protocol-level vulnerabilities from a defensive research perspective

Automation Request Portal (Microsoft Power Platform)

- Self-service application enabling users to browse, select and request automation playbooks through a multi-screen workflow with dynamic forms, filtered galleries, auto-generated request IDs and file attachments
- Architected five integrated Power Automate flows handling request validation, email notifications, CSV-to-SharePoint bulk import, attachment routing to document libraries, and structured data capture exported as a refreshable query
- Implemented an in-application work-notes and comments feature recording reviewer, timestamp, decision and remarks to build an auditable approval history; replaced a manual, phone-based approval process and reduced processing time by approximately 50%

Automation Catalogue & Access Provisioning Portal (Microsoft Power Platform)

- Searchable catalogue application allowing teams to discover and select automation solutions, with multi-filter search, data export, administrative CRUD interface and role-based access separating administrator and user views
- Built automated access provisioning using HTTP-based Microsoft Graph API calls — an access request triggers a form and Teams-based approval, after which Graph API queries add the approved user to the security group and grant application and document-library permissions
- Integrated Power BI dashboards to report automation deployment counts by region and account

Professional Experience

Senior Analyst — Automation Engineering | Capgemini India | January 2025 – Present

Capgemini is a global technology and consulting organisation delivering digital transformation, automation and enterprise IT solutions across industries.

- **Working** on the deployment of a distributed Ansible Automation Platform (AAP) integrating ServiceNow (ITSM), Splunk (observability), iPaaS and secrets management, enabling closed-loop, event-driven automation across customer environments
- **Performing** prerequisite architecture validation — operating systems, port matrix, service accounts, file partitions and component dependencies — across distributed nodes prior to deployment
- **Built** enterprise low-code applications and workflow automation using Microsoft Power Platform, integrating SharePoint, ServiceNow and Power BI; implemented access provisioning via HTTP-based Microsoft Graph API calls managing security-group membership and permissions
- **Integrated** Python into automation workflows for advanced data processing and validation
- **Collaborated** with international stakeholders across the USA, France, Poland and India on requirements and solution design

Analyst — Data & Automation Programs | Capgemini India | May 2023 – December 2024

- **Supported** analytics, tracking and governance for 118+ enterprise automation solutions
- **Developed** data-driven dashboards and reporting systems using Power BI and ServiceNow to measure automation benefits and operational performance
- **Built** low-code applications to improve automation visibility, governance and lifecycle management
- **Automated** repetitive operational and reporting tasks using Power Automate and Python, reducing manual effort and improving process efficiency
- **Gained** foundational experience across networking, Linux and Windows Server environments, system configuration and access control during onboarding and infrastructure training

Production Engineer — Operations | Basantar Breweries India | August 2020 – July 2022

Basantar Breweries is a manufacturing unit involved in large-scale brewing, bottling and canning operations.

- **Managed** production planning and operational workflows in a large-scale manufacturing setup; developed Excel-based tracking systems and a basic Android application for production monitoring and loss analysis

Freelance Subject Matter Expert — Computer Science | Chegg India | March 2020 – November 2021

- **Guided** students on web development fundamentals, SQL database queries and algorithmic problem-solving, maintaining high standards of accuracy and academic quality in a remote environment

Technical Skills

Systems & Automation: Ansible, Ansible Automation Platform (AAP), Linux (RHEL), Windows Server, systemd, Podman/containers, distributed deployment

Programming: Python (automation, static analysis, scripting), SQL

Infrastructure & Cloud: Microsoft Azure, ServiceNow, Splunk, Microsoft Graph API, REST APIs, Git/GitHub, CI/CD concepts

Concepts: Static analysis, idempotency, event-driven automation, closed-loop remediation, computer networks, systems security

AI & Low-Code: Agentic AI and LLM integration (n8n, LLM APIs, Ollama), Microsoft Power Platform (PowerApps, Power Automate), Power BI

Certifications

- Microsoft Certified: Azure Security Engineer (AZ-500)
- Microsoft Certified: Azure AI Fundamentals (AI-900)
- Microsoft Certified: Security, Compliance & Identity Fundamentals (SC-900)
- Microsoft Certified: Power Platform Functional Consultant (PL-200)
- Microsoft Certified: Power Platform Fundamentals (PL-900)
- Red Hat Certified Engineer (RHCE) EX294 — In Progress

Languages

- English (Fluent) • Hindi (Fluent)